BEHIND THE FEED · PHASE 1 VISUAL DATA STORY

What gets shown — and what gets overlooked?

We traced **8.6 million** logged article impressions from one week of US Microsoft News (MSN) back to the **65,238** articles in the dataset. What users were shown did not mirror what was available: **entertainment** received 5.1× its share of the catalogue, while **sports** — 30% of all articles — received 0.37×.

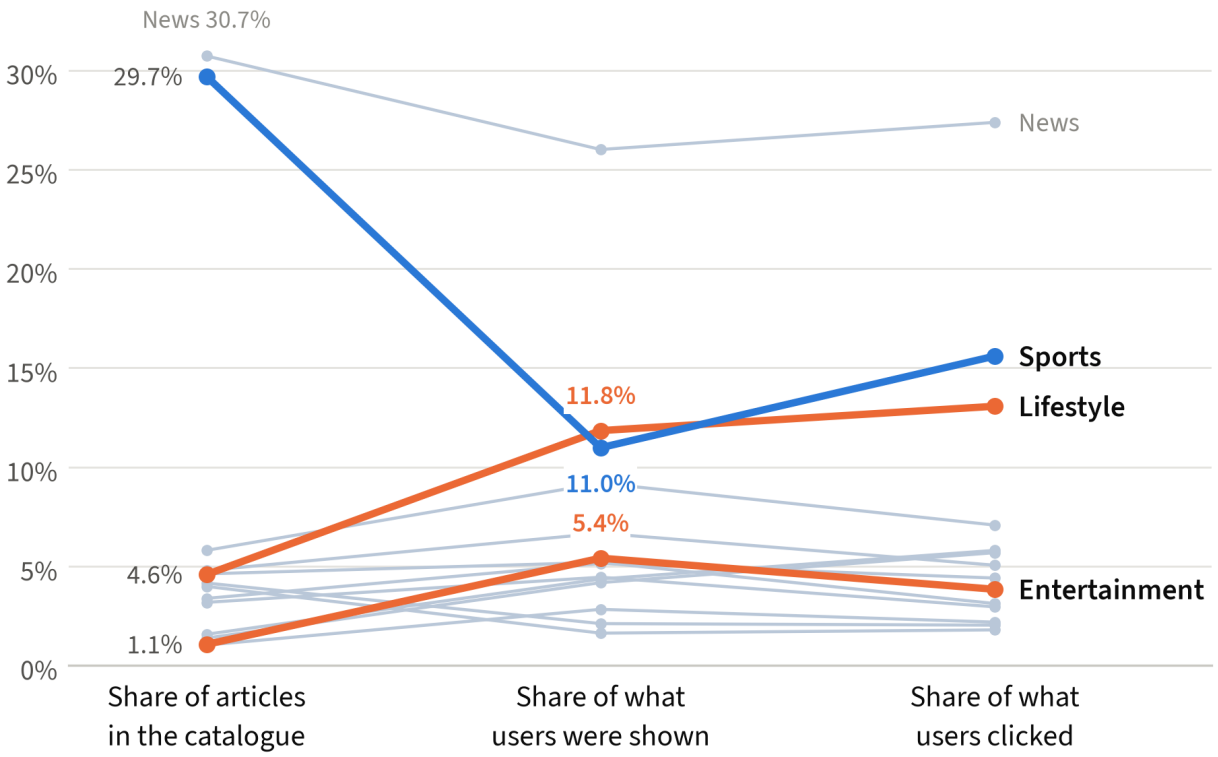
30%
of articles were sports…

11%
…but only this share of what users were shown

5.1×
Entertainment: exposure relative to its catalogue share

12%
Lifestyle: share of exposure, from 4.6% of articles

Share of each news category: in the catalogue → in what users were shown → in what they clicked



Each line is one of the 14 MIND categories with at least 50 articles. Orange: shown more than its catalogue share; blue: shown less; grey: other categories.

Sports was clicked most often when shown. Its click-through rate — clicks ÷ times shown — was 5.75%, the highest of any large category, yet it received the least exposure relative to the catalogue (0.37×; 0.39× against articles actually shown, second-lowest on that measure). Entertainment had one of the lowest click-through rates (2.88%) and the most amplified exposure.

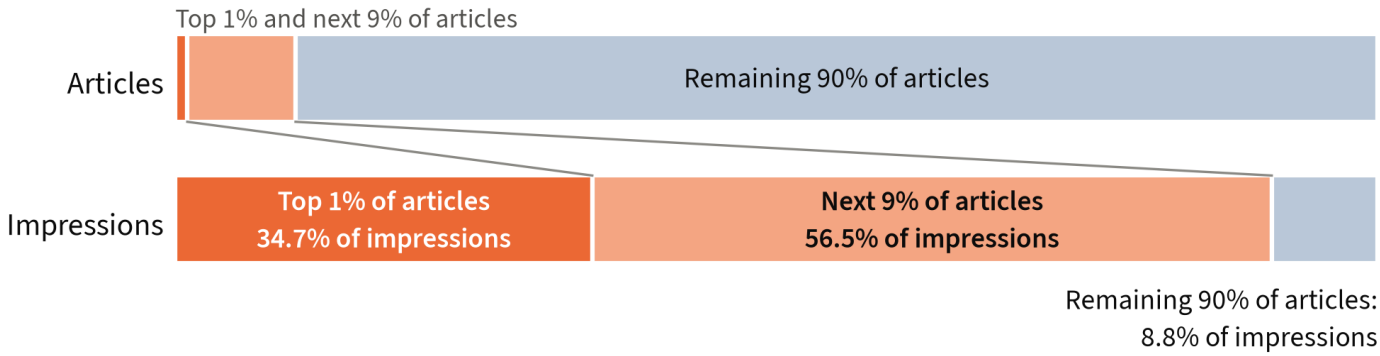
What this is — and isn't. An *impression* means an article was logged as shown; it is a proxy for exposure, not proof of reading. The pattern describes the output of Microsoft News's whole system — editors, page layout, recommendation and user navigation — not the recommender alone, and it does not show intent or unfairness.

Our question: does personalized information exposure come at the cost of diversity? News feeds decide which stories millions of people encounter. If what is shown drifts away from what is available, or narrows toward what each person already clicks, people see a thinner slice of the news without knowing it. Pages 2–3 measure this in the logged feed; page 4 tests, in simulation, whether a feed can be relevant *and* diverse.

WHO GETS THE SPOTLIGHT?

35% of all exposure went to 1% of articles

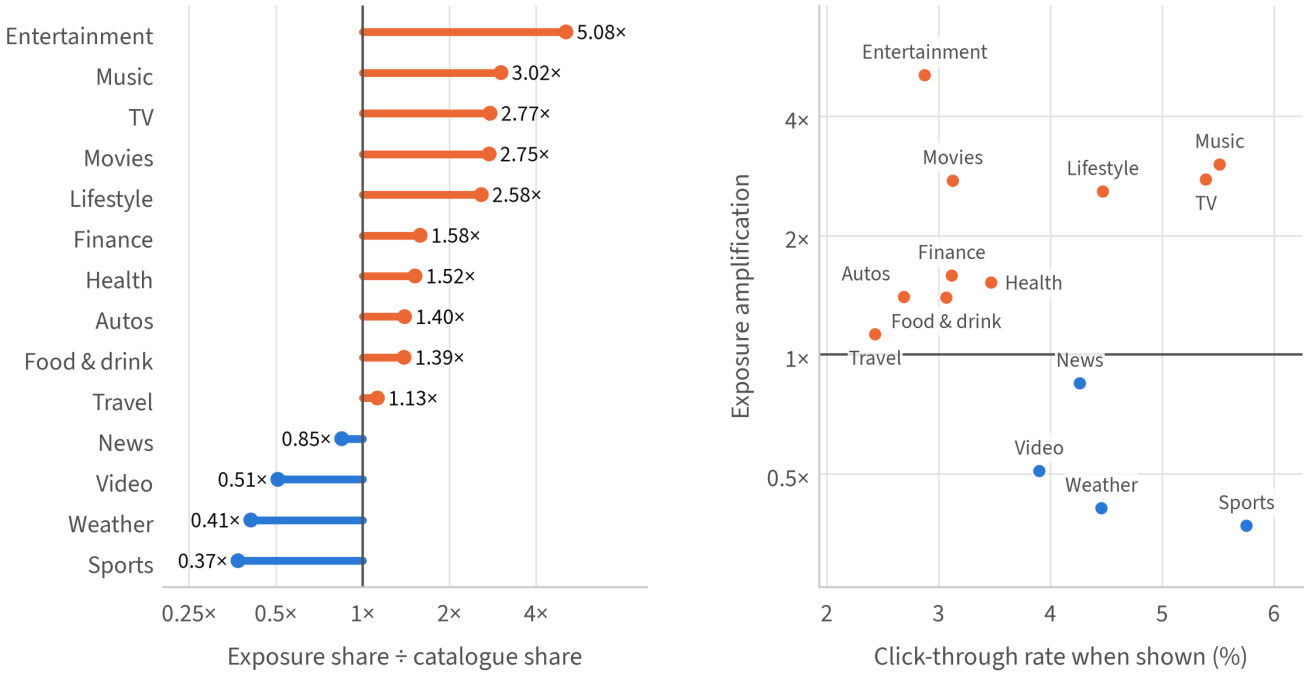
Of the 22,771 articles shown at least once, the most-shown 228 received 34.7% of all impressions and the top 10% received 91.2%. The bottom half of articles shared 0.4%.



Gini coefficient of impressions per article: 0.92 (0 = every article shown equally, 1 = one article shown everywhere). Within single days it ranges from 0.89 to 0.93, so the concentration is not just older and newer articles being mixed together.

Over-exposure is unrelated to click-through rate

Categories ranked by exposure relative to their catalogue share (left), and the same ratio against how often each category was clicked when shown (right).



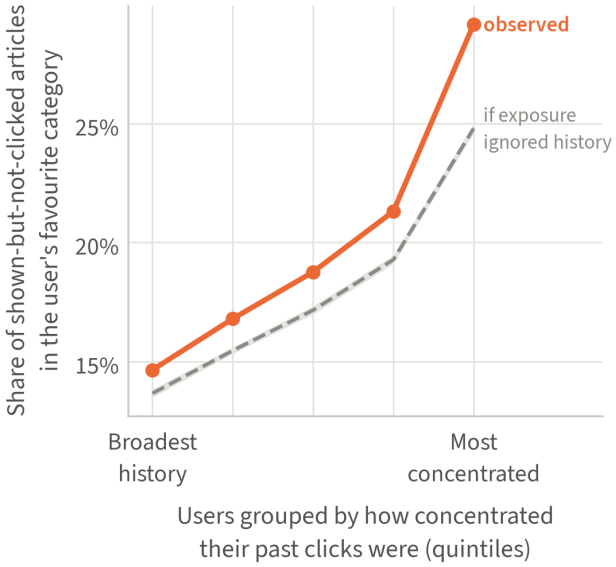
Across the 14 large categories, the rank correlation between exposure amplification and click-through rate is ρ = −0.08 (p = 0.78): no detectable relationship. Click-through rates are only comparable inside this dataset: every logged impression contains at least one click, so they describe engaged sessions. Four categories with fewer than 50 articles are excluded.

THE SHRINKING WINDOW?

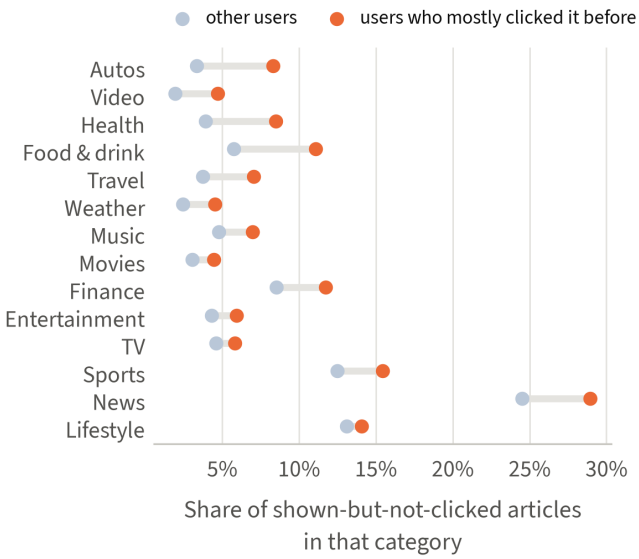
Past favourites are over-represented in what users see

For 77,216 users with at least 5 clicks *before* the logged week, their most-clicked category made up **20.1%** of the articles they were shown but did not click *during* it — versus an estimated 18.1% under a history-blind baseline (a 1.11× tilt; 1.15× counting every article shown). The tilt was largest for users whose past clicks were most concentrated (1.18× vs 1.07× for the broadest).

Exposure to each user's favourite category

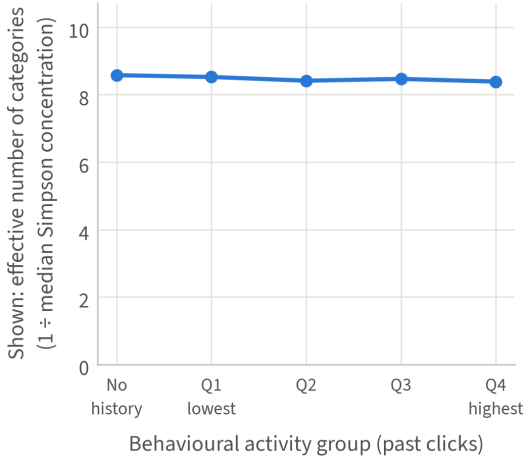


Past fans of a category see more of it than others



But exposure stays broader than clicks

At the median, a user's past clicks covered the equivalent of **5.5** evenly-used categories; what they were shown covered **8.4**. For 75% of users, past clicks were more concentrated than later exposure. Users with longer click histories were shown only negligibly more concentrated feeds (rank correlation between history length and exposure concentration ρ = 0.03).



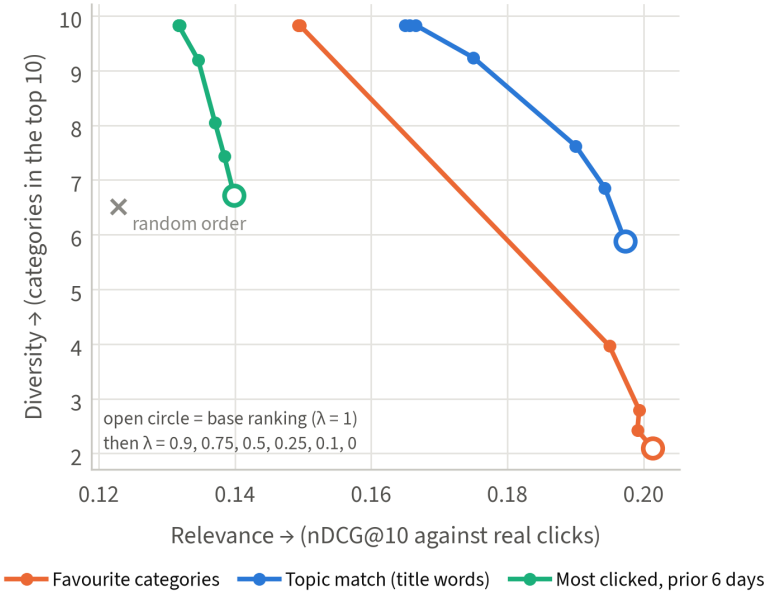
Associative design: history (clicks before the logs) precedes logged exposure (9–15 Nov 2019) but has no timestamps; this cannot show that past clicks *caused* later exposure. Logged impressions all contain a click, which favours users' interests, so the tilt is measured on articles shown but *not* clicked (a conservative choice). "Effective categories" = 1 ÷ the median chance that two random items share a category. Groups are behavioural, never demographic.

RELEVANCE VS. DIVERSITY — A SIMULATION

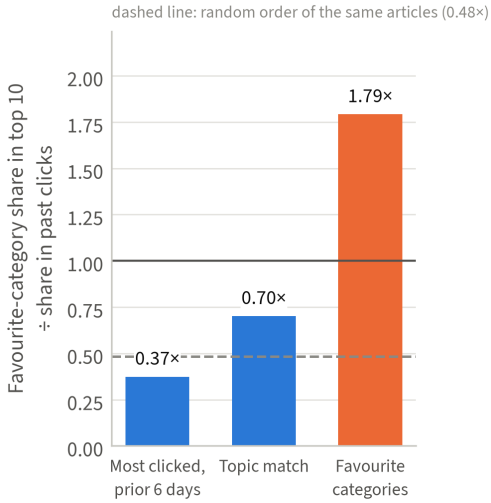
A little diversity costs very little relevance

We built three simple, transparent recommenders and re-ranked each one with **Maximal Marginal Relevance** (MMR), which trades predicted relevance against similarity to articles already picked. For the most accurate strategy (favourite-category, by a narrow margin over topic-match: nDCG@10 0.201 vs 0.197; AUC 0.606 vs 0.571), moving the dial to λ = 0.5 changed the number of categories in a top-10 list by **+89%** (2.1 → 4.0) while relevance changed by **−3.2%**.

Each dot is one setting of the diversity dial



Do recommenders amplify favourites?



The most accurate strategy was also the narrowest. Ranked by past behaviour, the favourite-category strategy put 2.1 categories in a top 10 (random order: 6.5) and filled it with the user's favourite category at 1.79× its share in their own history; topic-matching did so at 0.70×, a random order at 0.48×. Above 1× means the list is *narrower* than the user's own past clicks.

What we recommend. (1) Platforms should publish routine exposure audits like pages 1–3 — catalogue vs. exposure share, concentration, and tilt toward past clicks. (2) In simulation the diversity dial was cheap and explainable, so for Phase 2 we propose prototyping a user-facing “breadth” control that shows its relevance cost openly — and testing it with people.

Simulated strategies — they do not reproduce MSN's recommender. Evaluated on 39,581 real dev-day impressions (≥20 shown articles): re-ranking only the articles actually shown, scored against real clicks (nDCG@10). Models used only information available beforehand (pre-period histories, clicks from the six preceding days); 34% of test-day candidate articles had any click in those days.

So — does personalized exposure cost diversity?

In the observed feed, a modest lean. Among articles shown but not clicked, users saw 1.11× as much of their past favourite category as a history-blind baseline predicts — yet what they were shown stayed broader than what they clicked (8.4 vs 5.5 effective categories).

In our simulation, the default narrows. The most accurate strategy put 2.1 categories in a top 10.

Breadth is cheap to add. In simulation, a diversity dial raised categories per top 10 by 89% for a 3.2% relevance cost.

METHODS & LIMITATIONS

How we know — and what we can't conclude

Data. Microsoft News Dataset (MIND-small), Microsoft Research: 65,238 news articles with categories, and anonymized logs of 94,057 users — 230,117 impression logs (9–15 Nov 2019) containing 8,584,442 article impressions and 347,727 clicks.

Definitions. Exposure = an article logged as shown in an impression (a proxy, not proof of reading). **Engagement** = a click. **Catalogue share** = articles in a category ÷ all articles; **exposure share** = impressions ÷ all impressions; **CTR** = clicks ÷ impressions in that category.

Amplification. Exposure share ÷ catalogue share (1× = proportional). Because 65% of catalogue articles appear only in users' older click histories, we repeated it against articles actually shown at least once: the category ranking barely changes (rank correlation 0.97).

Concentration. Top-1%/10% shares and the Gini coefficient of impressions per article, also computed day by day.

Narrowing. Unbiased Simpson concentration (chance two items share a category), entropy, and a permutation test on articles shown but not clicked (logged impressions all contain a click, which favours interests): users' favourite categories are shuffled within groups of similar activity and period to estimate exposure if it ignored history. The tilt above the null held in 8 of 8 alternative specifications (history thresholds, train/dev-only exposure, activity strata).

Recommenders. Most clicked (clicks in the six days before the test day), TF-IDF title similarity to the user's history, and favourite-category preference; MMR re-ranking with λ from 1 to 0. Evaluated on the dev day with nDCG@10, AUC, distinct categories, intra-list diversity, coverage and an amplification ratio. No dev clicks used as input.

Limitations

- Observational data: associations, not causes.
- One week of US Microsoft News; not the whole internet, other platforms, Canada, or other years.
- Users are anonymized; no demographic or location data exist, and none were inferred. Groups are behavioural.
- Impressions are an exposure proxy; they do not prove an article was seen or read.
- Every logged impression contains at least one click, so click-through rates describe engaged sessions and are comparable only within this dataset.
- Click histories pre-date the logs but have no timestamps and never update during the logged week.
- Observed exposure reflects MSN's whole ecosystem — editors, layout, recommendation, navigation — not one algorithm.
- Our recommenders are simulations on logged candidate lists; they do not reproduce MSN's proprietary system.
- Diversity is measured with 18 broad categories; viewpoint and source diversity are not measured. Title-word diversity (ILD) was near its maximum for every strategy and is not used for conclusions.
- MIND-small is a sample of MIND users (50,000 per split), not a census of MSN traffic.

Sources. Wu, F. et al. (2020). MIND: A Large-scale Dataset for News Recommendation. *Proceedings of ACL 2020*, 3597–3606. Dataset: Microsoft News Dataset (MIND), Microsoft Research — https://msnews.github.io/ (Microsoft Research License Terms). Carbonell, J. & Goldstein, J. (1998). The use of MMR, diversity-based reranking for reordering documents and producing summaries. *SIGIR '98*. **Reproducibility.** All code, notebooks and the result files behind every number: https://github.com/suryalionael/information-diversity. The dataset itself is not redistributed.